NASA SPACE APPS CHALLENGE 2026 · ARCHITECTURAL BLUEPRINT

ApolloHarness Lunar Analytics

Complete architectural topology, real-time geometric interrogation lifecycle, and mission planning decision trees for lunar South Pole site evaluation.

DOCUMENT PLATES & ARCHITECTURE MAP

Plate 01: Full System C4 Architecture — Plate 01
Firebase Presentation, Render Docker Backend, Compute Core & Data LFS

Plate 02: Real-Time Execution Sequence — Plate 02
IM-2 Site Interrogation: SPICE Ephemeris, LOLA DEM Masks & 3D HUD

Plate 03: Core Mission Use Cases & Loops — Plate 03
Automated Solar Hard Gate, DTE Blackout Prediction & PSR Drill Standoff

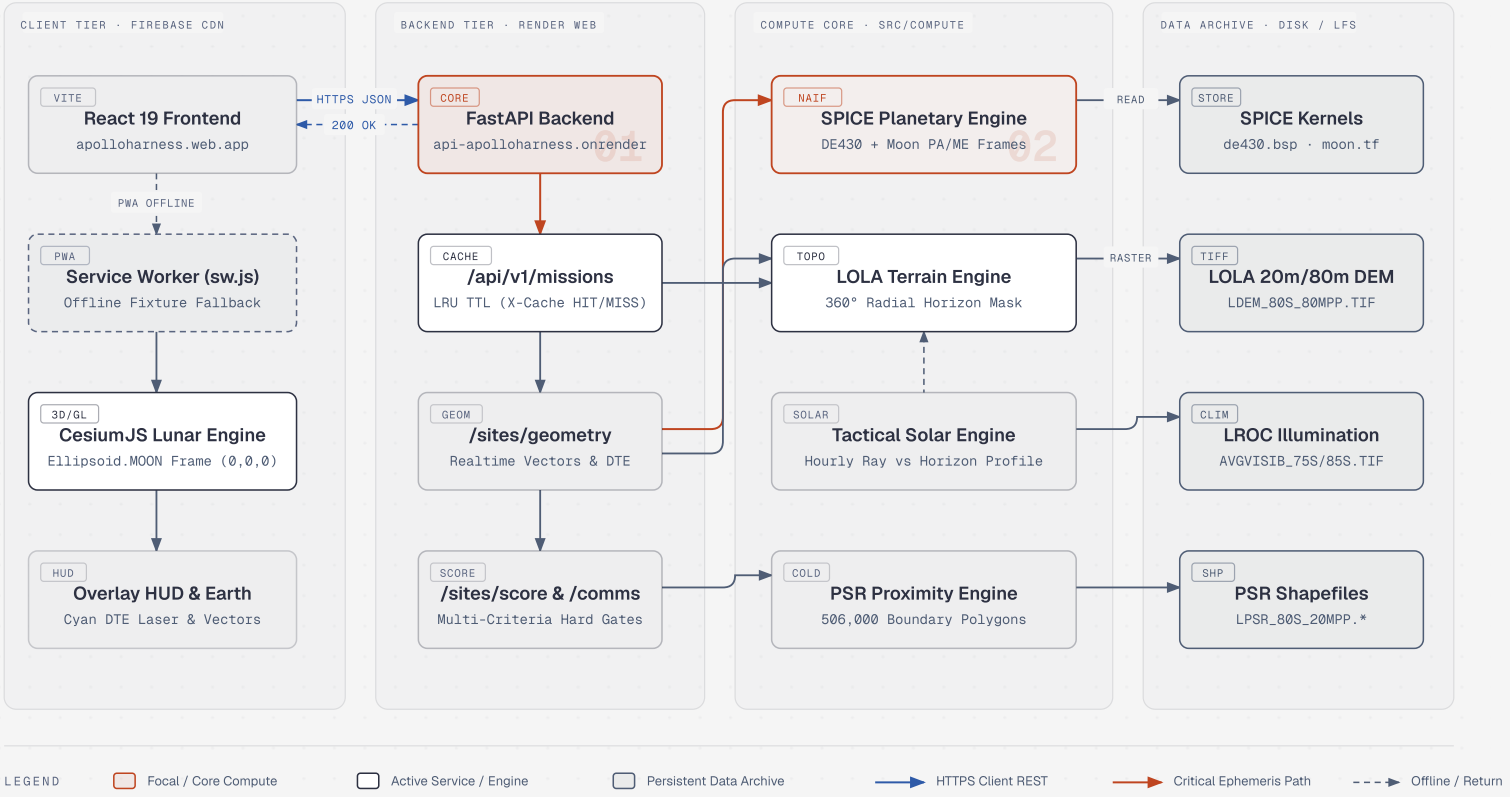
PLANETARY COMPUTE & DATASET PROVENANCE

CELESTIAL EPHEMERIS	NASA NAIF SPICE de430.bsp + moon_pa/me
LUNAR TOPOGRAPHY	LOLA 20m/80m DEM GeoTIFF (LDEM_80S_80MPP)
DECENNIEL ILLUMINATION	LROC/LOLA PGDA Product 69 (18.6-yr cycle)
CRYOGENIC COLD TRAPS	506,349 Permanently Shadowed Regions (<110K)
CLIENT & BACKEND CLOUD	Firebase CDN (SPA + PWA) + Render (Python 3.11)
DESIGN SYSTEM	cathrynlavery/diagram-design v2.6 (WCAG 2.1)

PLATE 01 · SYSTEM TOPOLOGY

Full System C4 Architecture Topology

Client presentation deployed to Firebase CDN, containerized FastAPI compute orchestrator on Render, and scientific compute core executing SPICE ephemerides and LOLA DEM horizon masks.



Scientific Pipeline & Ephemeris

NASA NAIF SPICE kernel integration provides sub-arcsecond celestial vectors. LOLA 20m/80m DEM extracts a full 360° radial horizon profile in <140ms, testing whether mountain rims occlude direct line-of-sight to Earth or the Sun.

Application Routing & Caching

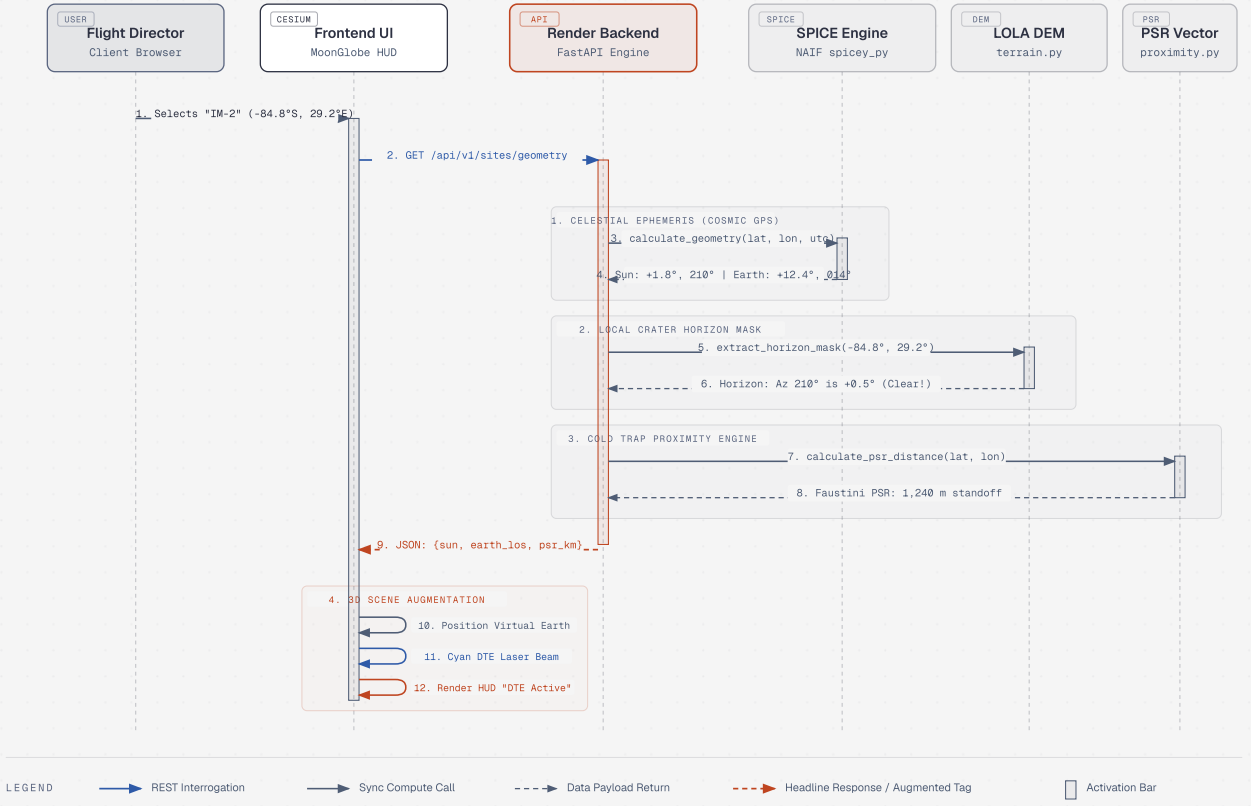
FastAPI backend running Python 3.11 Docker on Render with in-memory LRU TTL caching (`X-Cache HIT/MISS`). Direct query responses average <45ms for cached geometry evaluations.

Presentation & Offline PWA

React 19 + CesiumJS lunar globe hosted on global Firebase CDN. Registered Service Worker (`sw.js`) provides instant offline fixture fallback if live network drops during presentations or field audits.

Real-Time Geometric Interrogation Sequence

Complete trace of a landing site evaluation for IM-2 (-84.80°S, 29.20°E): from flight director selection to SPICE celestial vectors, LOLA horizon mask ray-tracing, and 3D Cesium HUD scene augmentation.



Sub-Arcsecond Celestial Ephemeris

SPICE calculates the precise planetary vectors at epoch `2026-11-15T12:00:00Z`. Solar elevation is +1.8° at azimuth 210°, while Earth hovers at +12.4° elevation at azimuth 014°.

Radial Terrain Occlusion Check

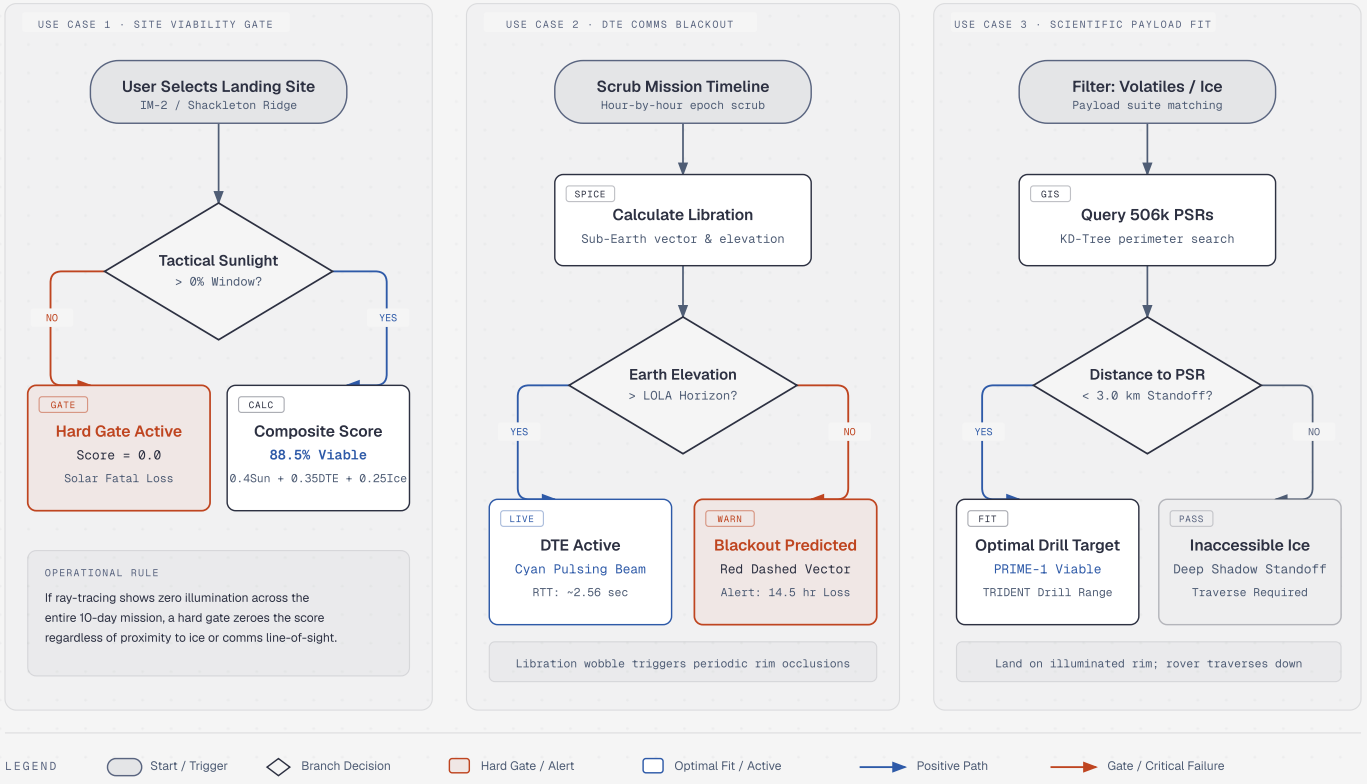
LOLA DEM 360° radial profiling confirms crater rims at azimuth 210° rise only to +0.5° elevation, confirming unshaded solar line-of-sight and direct RF link clearance to Earth.

Target Standoff & Scene Augmentation

PSR distance calculation determines a 1,240m standoff to the Faustini cold trap. The Cesium engine renders the cyan pulsating DTE laser beam directly to the virtual Earth sphere in Moon-centric coordinates.

Core Mission Use Cases & Decision Loops

Operational decision trees executing during flight planning: automated solar hard-gate viability enforcement, Direct-To-Earth (DTE) comms blackout prediction, and volatile payload science matching.



1. Automated Solar Hard Gate

Landers without RTGs cannot survive inside permanent shadow. If LOLA ray-tracing detects 0% contiguous sunlight across the mission window, the site is immediately zeroed to protect mission planning from fatal thermal and power failure.

2. Libration Comms Blackouts

Because the Moon's polar axis librates by ±6.7°, Earth dips below local mountain skylines even if average elevation is positive. The system precomputes line-of-sight across 240 hours to warn flight directors of expected multi-hour radio silences.

3. Scientific Payload Targeting

Volatile sampling payloads like NASA PRIME-1 / TRIDENT drill require close standoff (<3.0 km) to permanently shadowed craters. The engine verifies proximity to 506,000 cold trap perimeters to maximize science return.